Artyom Tuzov

ML Researcher
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Summary

I work in machine learning and robotics and at the intersection of the two. I run full-cycle applied research: from reading papers and reproducing results to a working demo handed over to a product team. Topics: NLP, RAG and search systems, agentic systems, computer vision, robot control on ROS2. Stack: Python, C++, PyTorch.

Education

Innopolis University Sept 2023 – June 2027

BSc, Data Science and Artificial Intelligence, robotics track. GPA 4.83

Experience

MWS RnD June 2026 – present

Research Engineer

- Full-cycle applied research: picking a topic, reproducing published results, building a working demo, finding an internal owner and handing the solution over
- Reworked hybrid search in a running project, metrics went up from a baseline level to close to SOTA
- Domains: NLP, RAG, search systems, agentic systems, harnesses

Innopolis University Sept 2025 – June 2026

Robotics Software Engineer

- **LLM swarm:** ROS 2 Humble, Stage, Nav2. PBS MAPF planner in C++ with gradient inflation and capsule-based conflict detection; LLM advisory layer over BehaviorTree with three channels: reactive, proactive and operator chat
- **PCB assembly inspection:** pcb_inspector ROS2 node: component detection and geometry report against a baseline; 12-class dataset, YOLO11s up to mAP50 0.964 at 15.1 ms on RTX 3050m
- **Unitree A1 robot dog:** ROS2 perception pipeline: YOLO11 pose via ONNX Runtime CUDA, IoU tracking, tuned for Jetson Orin Nano 4GB up to 15 FPS end-to-end

Yandex Cloud Dec 2025 – Mar 2026

ML Engineer

- Built an image classification service trained on soft labels: dataset assembly and balancing, CNN backbone benchmark, release. The service shipped to production
- Worked with noisy soft labels on a review scoring task: hunted labelling anomalies and kept QWK under control under heavy class imbalance

Innopolis University July 2026 – Aug 2026

Instructor, Professional Development Course

- Built the materials for a 72-hour course "AI Systems Specialist" from scratch for 20 students: applied Python and data work, generative models, fine-tuning and services on top of them
- Taught on site and over video, ran the final assessment

Awards

1st place, Russian Championship in Robotics Programming Dec 2025

- Russian Federation of Sport Programming. Finals: five applied automation tasks on Promobot M Edu manipulators, 13 teams from 8 regions

2nd place, Tenderhack Mar 2026

- Matching comparable goods for initial procurement price estimation: BM25, embeddings, ranking

Projects

ViKi May 2026 – present

- Diploma research project: a vision-to-kinematics pipeline for imitation learning on a robot manipulator
- Multi-camera RGB-D capture of demonstrations, pose estimation and retargeting onto robot kinematics

Cinematic Navigation from 3D Gaussian Splatting Mar 2026

- GPU pipeline for flying through a 3DGS scene: two camera planners, YOLO detection back-projected into world coordinates, revisit pass

Leadership

- **Innopolis Robotics Society**, co-founder of a student robotics community: we put teams together for hackathons and championships, run projects and mentor first-year students
- Young && Yandex ambassador, top 60 out of 7000+ applications; organised an internal hackathon for 70+ participants

Skills

ML / NLP: PyTorch, LLM, VLM, RAG, hybrid search, embeddings, ranking, fine-tuning

CV / Robotics: OpenCV, YOLO, ONNX Runtime, ROS2, Jetson, CUDA, Gazebo, MuJoCo, URDF, nav2

Systems & Data: Python, C++, Linux, Bash, Docker, Git, NumPy, Pandas, scikit-learn

Languages: Russian native, English B2